Taya Bhavsar

BUS235E

Data Visualization and Management

6 May 2026

Amazon E-Commerce Beauty Trends (Jan–Jun 2024)

I. Introduction

My objective for this analysis is to examine purchasing behavior in Amazon's beauty category, focusing specifically on the Makeup and Haircare subcategories. Using over 125,000 records from January to June 2024, this project investigates shifts in purchase volume, variations in discounts by subcategory, and the connection between discount levels and final prices. These findings reveal seasonal patterns, pricing dynamics, and category performance in the beauty segment.

The project is guided by four business questions:

1. How did purchase volume change from March to June 2024?
2. Do makeup and hair care receive different average discount levels?
3. How do discounts relate to the final price?
4. How did purchase patterns differ by subcategory and month?

To answer these questions, I cleaned, visualized, and analyzed the dataset using a variety of visualizations, including a calculated field that determines the interpretability of the relationship between discount levels and final prices.

II. Dataset Overview

The Amazon E-Commerce dataset records individual beauty product purchases. Key fields used in this analysis include Discount, Final Price, Rating, Subcategory, and Purchase Date. To maintain focus and reduce noise, the dataset was filtered to include only Makeup and Haircare purchases between January 1 and June 30, 2024.

Data preparation involved importing the dataset into Excel, filtering by category and date, and removing irrelevant rows from the original one-million-record file. This ensured a clean, manageable dataset for Tableau. Cleaning the dataset in Excel allows target specificity of the data to maintain accuracy and without any errors. Removing the extra rows in Excel allowed Tableau to import the dataset with no issues, improving performance.

In Tableau, I created a *Discount Tier* calculated field that groups discount percentages into Low (<15%), Medium (15–30%), and High (≥30%). This field enhances the scatter plot by grouping discount levels into meaningful categories. Final Price was aggregated using *Average* to maintain consistency, and tooltips were adjusted to avoid misleading aggregations such as summed prices or ratings.

III. Visualization Design Choices

Each visualization in this analysis was intentionally selected to address a specific business question and to present the data in a clear, intuitive way. The line chart illustrating purchase volume from March through June was chosen because it effectively highlights month-to-month changes and makes broader seasonal trends immediately visible. The sharp increase from March to April stands out clearly, helping to contextualize shifts in consumer demand. To compare discount levels between Makeup and Haircare, a bar chart was used due to its simplicity and directness; it clearly shows that both subcategories receive nearly identical average discounts, with Makeup only slightly higher. The scatter plot examining the relationship between discount percentage and average final price was selected because it is well-suited for identifying correlations. The downward trend in the plot demonstrates that higher discounts are associated with lower final prices, and the addition of the Discount Tier field enhances interpretability by grouping discount levels into meaningful categories. Finally, the highlight table was used to display purchase counts across subcategories and months. This format allows for quick comparison across two dimensions, and the use of shading makes it easy to identify peak months and stronger-performing subcategories at a glance.

IV. Beauty Trend Key Insights

The visualizations reveal several important insights about beauty purchasing behavior on Amazon during the first half of 2024. Purchase volume was extremely low in March, with fewer than 100 purchases per subcategory, but activity increased dramatically in April, May, and June, each of which recorded between 2,600 and 2,800 purchases. This sharp rise suggests a seasonal or promotional shift in consumer behavior during the second quarter of the year.

The comparison of discount levels shows that Makeup and Haircare receive nearly identical average discounts of around 25 percent, with Makeup only slightly higher. This consistency indicates that Amazon applies a uniform promotional strategy across these beauty categories. The scatter plot further demonstrates a clear relationship between discount percentage and final price: as discounts increase, final prices decrease. High discounts in the 40 to 50 percent range correspond to significantly lower final prices, often between $500 and $800, confirming that discounting is an effective and predictable pricing lever.

The highlight table adds another layer of insight by showing that Makeup consistently records slightly higher purchase counts than Haircare during the high-activity months of April, May, and June. Despite receiving similar discount levels, Makeup drives more volume, which may reflect stronger consumer interest or a broader variety of available products. Across all visualizations, April and May emerge as the most active months, suggesting that seasonal promotions, product launches, or increased springtime interest in beauty products may have influenced purchasing behavior during this period.

V. Limitations

Although this analysis provides strong insights, several limitations should be acknowledged. The dataset includes only January through June 2024, so full-year trends cannot be assessed. The analysis focuses solely on Makeup and Haircare, excluding other beauty categories such as skincare, which may exhibit different purchasing behavior. The dataset does not include demographic or geographic information, limiting the ability to segment customers or identify regional patterns. Using average final price may obscure outliers or variations within specific product types. Additionally, discount data may not capture all promotional mechanisms, such as coupon usage or bundled offers.

VI. Conclusion

This analysis provides a clear understanding of beauty purchase trends on Amazon during the first half of 2024. Purchase volume increased sharply from March to June, with April and May representing the strongest months. Makeup and Haircare receive similar average discounts, yet Makeup slightly outperforms Haircare in purchase volume during peak months. The relationship between discount and final price is strong and predictable, with higher discounts consistently leading to lower final prices. Overall, the analysis successfully answers the four guiding business questions and provides a clear view of beauty purchasing behavior during early 2024.

Future analysis could expand the scope to include additional subcategories, full-year data, or customer segmentation. Incorporating demographic or geographic information would provide deeper insights into consumer behavior and help identify targeted promotional opportunities.

Dataset

SharmajiCoder. (2024, March). Amazon E-Commerce Dataset, Version 1. Retrieved April 23, 2026, from
https://www.kaggle.com/datasets/sharmajicoder/amazon-e-commerce